

COEFICIENTES ALINE

O kernel agora será

$$\begin{aligned} & \int_0^\infty d\gamma' \frac{g_i(\gamma', z)}{[\gamma + \gamma' + \kappa^2 + z^2 M^2/4]^2} \\ &= -\frac{\lambda_S \lambda_F}{16\pi^2} \sum_{j=1,2} \int_{-1}^{+1} dz' \int_0^\infty d\gamma' \left(\mathcal{P}_{ij}^{(1)}(\gamma, z, \gamma', z') + (1 - \xi) \mathcal{P}_{ij}^{(2)}(\gamma, z, \gamma', z') \right) g_j(\gamma', z'), \end{aligned} \quad (4.65)$$

$$\begin{aligned} \mathcal{P}_{ij}^{(2)}(\gamma, z, \gamma', z') &= -2 \int_0^1 v^2 dv \int_0^{1-v} du \\ & \left[(1+z)^3 \frac{\theta(z' - z)}{D_0(\gamma, z)} \frac{1}{\left[D_u^{(2)}(z', z, m_s^2) \right]^3} \frac{1}{(k_u^-)^2 + z M k_u^- + M^2 z^2/4 + 4\gamma} \times \right. \\ & \quad \left(d_{ij}^{(0)}(k_u^-) + f^{(2)}(k_u^-, \gamma', z'; u, v) \left(d_{ij}^{(2)}(k_u^-) + \frac{1}{4} d_{ij}^{(3)}(k_u^-) \left(\frac{M}{2} k_u^- - \frac{z M^2}{4} \right) \right. \right. \\ & \quad \left. \left. + \frac{3}{2} d_{ij}^{(4)}(k_u^-) f^{(2)}(k_u^-, \gamma', z'; u, v) + \frac{1}{4} d_{ij}^{(5)}(k_u^-) M^2 + \frac{1}{4} d_{ij}^{(6)}(k_u^-) \left(-\frac{Mz}{2} k_u^- - \gamma \right) \right) \right) \\ & + (1-z)^3 \frac{\theta(z - z')}{D_0(\gamma, z)} \frac{1}{\left[D_d^{(2)}(z', z, m_f^2) \right]^3} \frac{1}{(k_d^-)^2 + z M k_d^- + M^2 z^2/4 + 4\gamma} \times \\ & \quad \left(d_{ij}^{(0)}(k_d^-) + f^{(2)}(k_d^-, \gamma', z'; u, v) \left(d_{ij}^{(2)}(k_d^-) + \frac{1}{4} d_{ij}^{(3)}(k_d^-) \left(\frac{M}{2} k_d^- - \frac{z M^2}{4} \right) \right. \right. \\ & \quad \left. \left. + \frac{3}{2} d_{ij}^{(4)}(k_d^-) f^{(2)}(k_d^-, \gamma', z'; u, v) + \frac{1}{4} d_{ij}^{(5)}(k_d^-) M^2 + \frac{1}{4} d_{ij}^{(6)}(k_d^-) \left(-\frac{Mz}{2} k_d^- - \gamma \right) \right) \right) \left. \right] \end{aligned}$$

$$\begin{aligned} f^{(2)}(k_u^-, \gamma', z'; u, v) &= -\gamma \frac{v(1-v)(1+z')}{1+z} - v(\gamma' + \kappa^2) + \frac{M^2 v}{4} (-z(1+z')(1-v) + z'(1-v - vz')) \\ & \quad - m_s^2 \frac{v(1-v)(z' - z)}{1+z} - \mu^2(1-u-v+\xi u) \end{aligned}$$

$$\begin{aligned} f^{(2)}(k_d^-, \gamma', z'; u, v) &= -\gamma \frac{v(1-v)(1-z')}{1-z} - v(\gamma' + \kappa^2) + \frac{M^2 v}{4} (z(1-z')(1-v) - z'(1-v + vz')) \\ & \quad - m_f^2 \frac{v(1-v)(z - z')}{1-z} - \mu^2(1-u-v+\xi u) \end{aligned}$$

$$D_0(\gamma, z) = \gamma + (1 - z^2)\kappa^2 + (\Delta - z\bar{m})^2$$

$$l_D^{(2)} = l_D^{(1)} + \mu^2(1 - \xi)u.$$

$$k_D = \frac{M}{2}v(1-v)(z' - z),$$

$$l_D^{(1)} = -v(1-v)\left(\gamma + \frac{M^2}{4}zz'\right) - v\left(\kappa^2 + \frac{M^2}{4}z'^2v + \gamma'\right) - (1-v)\mu^2.$$

$$k_u^- = \frac{M}{2} - \frac{2(\gamma + m_S^2)}{M(1+z)}$$

$$k_d^- = -\frac{M}{2} + \frac{2(\gamma + m_F^2)}{M(1-z)}$$

$$D_u^{(1)}(z', z, m_s^2, m_f^2) = -(1+z)\left(k_D k_u^- + l_D^{(1)}\right)$$

$$D_d^{(1)}(z', z, m_f^2, m_s^2) = -(1-z)\left(k_D k_d^- + l_D^{(1)}\right)$$

- Coeficientes para $k^- = k_u^-$:

$$d_{11}^{(0)} = -\frac{v^2}{32(1+z)^2} \left[-4m_s^2(-1+z) + M^2(1+z)(-1+2z-z') - 2m_f M(1+z)z' + 8\gamma \right] \times \\ \left[M^2(1+z)(1+z')(-1+(-1+v)z-vz') - 4m_s^2(-1+(-2+v)z+z'-vz') \right. \\ \left. + 4(-1+v)(1+z')\gamma \right] \quad (C.5)$$

$$d_{12}^{(0)} = \frac{v^2}{64(1+z)^2} \left[M(1+z)(1+z')(-1+(-1+v)z-vz') - 4\frac{m_s^2}{M}(-1+(-2+v)z+z'-vz') \right. \\ \left. + \frac{4}{M}(-1+v)(1+z')\gamma \right] \left[2m_f M(1+z)(z'-(1+z')(z-vz+ vz')) + M^2(1+z)(-z'(1+z') \right. \\ \left. + Mz(-1+v+z'+vz')) + 8\frac{m_f}{M}(-1+v)(m_s^2(-z+z') + \gamma + z'\gamma) + 4(m_s^2(vz'-z(-1+v+z')) \right. \\ \left. + (-1+v+z'+vz')\gamma) \right] \quad (C.6)$$

$$d_{21}^{(0)} = -\frac{1}{(16(1+z))} \left[(Mv^2(2m_f + M(-1+z'))(-M^2(1+z)(1+z')(-1+(-1+v)z-vz') \right. \\ \left. + 4m_s^2(-1+(-2+v)z+z'-vz') - 4(-1+v)(1+z')\gamma) \right] \quad (C.7)$$

$$d_{22}^{(0)} = \frac{v^2}{32(1+z)^2} \left[\frac{1}{M^2}(1+z)(1+z')(-1+(-1+v)z-vz') - 4m_s^2(-1+(-2+v)z+z'-vz') \right. \\ \left. + 4(-1+v)(1+z')\gamma \right] \left[-2m_f M(1+z)z' + (1+z)(2z'-(1+z')(z-vz+ vz')) \right. \\ \left. + 4(-1+v)(m_s^2(-z+z') + \gamma + z'\gamma) \right] \quad (C.8)$$

$$d_{11}^{(2)} = \frac{v(-4m_s^2(-1+z) + M^2(1+z)(-1+2z-z') - 2Mm_f(1+z)z' + 8\gamma)}{8(1+z)} \quad (C.9)$$

$$d_{12}^{(2)} = \frac{v}{(16(1+z))} \left[4m_s^2(-1+z'-2vz'+z(-3+2v+z')) - 2m_f M(1+z)(-1-2(1+z') \times \right. \\ \left. (z-vz+ vz')) - M^2(1+z)(2z(-1+v+ vz') - (1+z')(1+2vz')) - 8(-1+v+ vz')\gamma \right. \\ \left. + \frac{8m_f}{M}(m_s^2(-1-3z+2vz+2z'-2vz') - 2(-1+v)(1+z')\gamma) \right] \quad (C.10)$$

$$d_{21}^{(2)} = -\frac{Mv}{4}(2m_f + M(-1+z')) \quad (C.11)$$

$$d_{22}^{(2)} = -\frac{v}{(8(1+z))} \left[(-2m_f M(1+z)z' + 4m_s^2(1+(3-2v)z + 2(-1+v)z')) \right. \\ \left. + M^2(1+z)(-1+z' - 2(1+z')(z-vz+ vz')) + 8(-1+v)(1+z')\gamma \right] \quad (\text{C.12})$$

$$d_{11}^{(3)} = \frac{1}{2(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2)} \left[4M^3m_f(1+z)^2(z(3+v(-2+z')) - vz') \right. \\ - M^4(1+z)^2((-3+2v)(1+z(-1+2z)) + 2v(-1+3z)z') + 16(-3+2v)(m_s^2(-1+z) - 2\gamma) \times \\ (m_s^2 + \gamma) + 4M^2(1+z)(m_s^2((-3+2v)(2+(-2+z)z) + 2v(-1+2z)z')) \\ + (-3+12z+v(2-8z-6z'))\gamma) + 16Mm_f(1+z)(m_s^2(-3z+2vz+vz')) \\ \left. + (3+v(-2+z'))\gamma \right] \quad (\text{C.13})$$

$$d_{12}^{(3)} = \frac{1}{4(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2)} \left[2M^3m_f(1+z)^2(1+(5-2z)z \right. \\ + 4v^2(1+z)^2z' - 2v(1+3z+(3+z(4+3z))z')) + M^4(1+z)^2(1+z(-1+10z) \\ + 4v^2(1+z)^2z' - 2v(1+z+2z'+7zz'+z^2(4+z')) + 32\frac{m_f}{M}(m_s^2 + \gamma)(m_s^2(1+3z+4v^2z' \\ - 2v(1+z+3z')) + 2(-1+v(-3+2v)z')\gamma) - 4M^2(1+z)(m_s^2(2+z(-4+15z) \\ + 2v(-2+z^2(-5+z') + 4(-1+v)z' + 4(-2+v)zz')) + \gamma + 2(v-10z+8vz \\ + v(7+2z-4v(1+z))z')\gamma) - 8Mm_f(1+z)(m_s^2(2+(8-3z)z + 2v(-2+z(-4+z-4z') - 6z') \\ + 8v^2(1+z)z') + (-5+6v+4z+4v(2+3z-2v(1+z))z')\gamma) + 16(m_s^4(1-3z+6z^2+4v^2z' \\ - 2v(1+2z'+z(-1+2z+z')) + m_s^2(5-15z+2v(-3+5z-(3-4v+z)z'))\gamma + 2(5+2v^2z' \\ - v(4+z'))\gamma^2) \left. \right] \quad (\text{C.14})$$

$$d_{21}^{(3)} = \frac{1}{M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2} \left[M(1+z)(2M^2m_f(1+z)(3-3z \right. \\ + 2v(-1+z+2z')) + M^3(1+z)(-3+9z+2v(1-3z+(-3+z)z')) + 8m_f(-3+2v)(m_s^2 + \gamma) \\ \left. + 4M(m_s^2(3-6z+2v(-1+2z+z')) + (9+2v(-3+z'))\gamma) \right] \quad (\text{C.15})$$

$$d_{22}^{(3)} = \frac{1}{(2(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2))} \left[-4M^3m_f(1+z)^2(z(3+v(-2+z')) - vz') + M^4(1+z)^2(1+11z-2(v+5vz+z^2)+2v(-4-3z(1+z)+2v(1+z)^2)z') \right. \\ \left. - 16Mm_f(1+z)(m_s^2((-3+2v)z+vz')+(3+v(-2+z'))\gamma) + 16(m_s^2+\gamma)(m_s^2(1+3z+4v^2z' - 2v(1+z+3z')) + 2(-1+v(-3+2v)z')\gamma) - 4M^2(1+z)(m_s^2(2+(14-3z)z+2v(-2+z(-6+z-4z')-7z')+8v^2(1+z)z') + (-11+4z+2v(5+(3+6z-4v(1+z))z'))\gamma) \right] \quad (\text{C.16})$$

$$d_{12}^{(4)} = \frac{M+2m_f}{4M} \quad (\text{C.17})$$

$$d_{22}^{(4)} = \frac{1}{2} \quad (\text{C.18})$$

$$d_{11}^{(5)} = -\frac{1}{(2M^4(1+z)^4 - 16M^2(1+z)^2(m_s^2 - \gamma) + 32(m_s^2 + \gamma)^2)} \left[(1+vz')(4M^3m_fz(1+z)^2 + M^4(1+z+z^2+3z^3+2z^4) - 16Mm_f(1+z)(m_s^2z - \gamma) - 16(m_s^2(-1+z) - 2\gamma)(m_s^2 + \gamma) - 4M^2(1+z)(m_s^2(2+(-2+z)z) + \gamma - 4z\gamma)) \right] \quad (\text{C.19})$$

$$d_{12}^{(5)} = \frac{(1+vz')}{(4(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2))} \left[(2M^3m_f(1+z)^2((-1+z)z + v(1+z)^2z') + M^4(1+z)^2(z-3z^2+v(1+z)^2z') - 32\frac{m_f}{M}(m_s^2+\gamma)(m_s^2(z-vz') - (1+vz')\gamma) + 4M^2(1+z)(m_s^2(z(-2+5z)-2v(1+z)z') + (1-6z+2v(1+z)z')\gamma) + 8Mm_f(1+z)(-m_s^2((-2+z)z+2v(1+z)z') + (-1+2z+2v(1+z)z')\gamma) + 16(m_s^4(z-2z^2+vz') + m_s^2(-1+5z+2vz')\gamma + (-3+vz')\gamma^2)) \right] \quad (\text{C.20})$$

$$d_{21}^{(5)} = -\frac{M(1+z)(1+vz')}{(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2)} \left[(M^3(1+z)(-1+3z) - 2M^2m_f(-1+z^2) - 8m_f(m_s^2 + \gamma) + 4M(m_s^2(1-2z)+3\gamma)) \right] \quad (\text{C.21})$$

$$d_{22}^{(5)} = \frac{(1 + vz')}{(2(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2))} [(4M^3m_fz(1+z)^2 + M^4(1+z)^2((-3+z)z + v(1+z)^2z') - 16Mm_f(1+z)(m_s^2z - \gamma) - 16(m_s^2 + \gamma)(m_s^2(z - vz') - (1 + vz')\gamma) - 4M^2(1+z)(m_s^2((-4+z)z + 2v(1+z)z') + 3\gamma - 2(z + v(1+z)z')\gamma))] \quad (\text{C.22})$$

$$d_{11}^{(6)} = -\frac{2M(-1+v)(1+z)}{(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2)} [(M^3(1+z)(-1+3z) - 2M^2m_f(-1+z^2) - 8m_f(m_s^2 + \gamma) + 4M(m_s^2(1-2z) + 3\gamma))] \quad (\text{C.23})$$

$$d_{12}^{(6)} = \frac{(-1+v)}{(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2)} [(M^4(1+z)^2((-3+z)z + v(1+z)^2) + 2M^3m_f(1+z)^2(-1-z^2 + v(1+z)^2) + 32\frac{m_f}{M}(-1+v)(m_s^2 + \gamma)^2 + 16(m_s^2 + \gamma)(m_s^2(v-z) + \gamma + v\gamma) + 16Mm_f(1+z)(-m_s^2(-1+v+ vz) + (v + (-1+v)z)\gamma) + 4M^2(1+z)(-m_s^2((-4+z)z + 2v(1+z)) - 3\gamma + 2(v+z+ vz)\gamma))] \quad (\text{C.24})$$

$$d_{21}^{(6)} = \frac{4M^2(-1+v)(1+z)[M(4m_f + M(-3+z))(1+z) + 4(m_s^2 + \gamma)]}{M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2} \quad (\text{C.25})$$

$$d_{22}^{(6)} = \frac{2(-1+v)}{(M^4(1+z)^4 - 8M^2(1+z)^2(m_s^2 - \gamma) + 16(m_s^2 + \gamma)^2)} [(-2M^3m_f(-1+z)(1+z)^2 + M^4(1+z)^2(-2+z-z^2 + v(1+z)^2) - 8Mm_f(1+z)(m_s^2 + \gamma) + 16(-1+v)(m_s^2 + \gamma)^2 + 4M^2(1+z)(m_s^2(3-2v(1+z)) + \gamma + 2(v + (-1+v)z)\gamma))] \quad (\text{C.26})$$

- Coeficientes em $k^- = k_d^-$:

$$d_{11}^{(0)} = -\frac{(M+2m_f)v^2(2m_f+M(-1+z'))}{32(-1+z)} [(4m_f^2(-1+(-2+v)z+z'-vz')-M^2(-1+z)\times \\ (1+z(3-v+(-1+v)z')+z'(-3+v-vz'))+4(-3+v+z'-vz')\gamma)] \quad (\text{C.27})$$

$$d_{12}^{(0)} = \frac{(M+2m_f)v^2}{(64(-1+z)^2)} \left[-4\frac{m_f^2}{M}(-1+(-2+v)z+z'-vz')+M(-1+z)(1+z(3-v+(-1+v)z') \right. \\ \left. +z'(-3+v-vz'))+\frac{4}{M}(3-v+(-1+v)z')\gamma \right] [-2m_f(-1+z)z'+M(-1+z)(-1+z')\times \\ ((-1+v)z-vz')+4(-1+v)(m_f^2(-z+z')+(-1+z')\gamma)] \quad (\text{C.28})$$

$$d_{21}^{(0)} = -\frac{Mv^2}{16(-1+z)} \{ [2m_f+M(-1+z')][4m_f^2(-1+(-2+v)z+z'-vz')-M^2(-1+z)\times \\ (1+z(3-v+(-1+v)z')+z'(-3+v-vz'))+4(-3+v+z'-vz')\gamma] \} \quad (\text{C.29})$$

$$d_{22}^{(0)} = \frac{v^2}{32(-1+z)^2} [-4m_f^2(-1+(-2+v)z+z'-vz')+M^2(-1+z)(1+z(3-v+(-1+v)z') \\ +z'(-3+v-vz'))+4(3-v+(-1+v)z')\gamma] [-2Mm_f(-1+z)z'+M^2(-1+z)(-1+z')\times \\ ((-1+v)z-vz')-4(-1+v)(m_f^2(z-z')+\gamma-z'\gamma)] \quad (\text{C.30})$$

$$d_{11}^{(2)} = -\frac{v}{8}(M+2m_f)(2m_f+M(-1+z')) \quad (\text{C.31})$$

$$d_{12}^{(2)} = -\frac{v(M+2m_f)}{16M(-1+z)} (-2Mm_f(-1+z)z'+4m_f^2(1+(3-2v)z+2(-1+v)z') \\ +M^2(-1+z)(1+2z(2-v+(-1+v)z')+z'(-3+2v-2vz')) \\ +8(2+v(-1+z')-z')\gamma) \quad (\text{C.32})$$

$$d_{21}^{(2)} = -\frac{Mv}{4}[2m_f+M(-1+z')] \quad (\text{C.33})$$

$$d_{22}^{(2)} = \frac{1}{(8(-1+z))} [v(2Mm_f(-1+z)z'+4m_f^2(-1-3z+2vz+2z'-2vz') \\ -M^2(-1+z)(1+2z(2-v+(-1+v)z')+z'(-3+2v-2vz')) \\ +8(-2+v+z'-vz')\gamma)] \quad (\text{C.34})$$

$$d_{11}^{(3)} = \frac{(M + 2m_f)(-1 + z)((M + 2m_f)(-3 + 2v) + 2Mvz')}{2(M + 2m_f - Mz)^2 + 8\gamma} \quad (\text{C.35})$$

$$\begin{aligned} d_{12}^{(3)} = & \frac{(M + 2m_f)}{(4M((M + 2m_f - Mz)^2 + 4\gamma))} (-4Mm_f(-1 + z)(1 - 2v + v(-5 + 4v)z') \\ & + 4m_f^2(1 + 3z + 4v^2z' - 2v(1 + z + 3z')) + M^2(-1 + z)(-1 + 4z \\ & + 2v(1 - 2z + (2 + 2v(-1 + z) - 3z)z')) + 8(2 + v(-2 + (-3 + 2v)z'))\gamma) \end{aligned} \quad (\text{C.36})$$

$$d_{21}^{(3)} = \frac{M(-1 + z)((M + 2m_f)(-3 + 2v) + 2Mvz')}{(M + 2m_f - Mz)^2 + 4\gamma} \quad (\text{C.37})$$

$$\begin{aligned} d_{22}^{(3)} = & \frac{1}{(2(M + 2m_f - Mz)^2 + 8\gamma)} [-4Mm_f(-1 + z)(1 - 2v + v(-5 + 4v)z') \\ & + 4m_f^2(1 + 3z + 4v^2z' - 2v(1 + z + 3z')) + M^2(-1 + z)(-1 + 4z + 2v(1 - 2z \\ & + (2 + 2v(-1 + z) - 3z)z')) + 8(2 + v(-2 + (-3 + 2v)z'))\gamma] \end{aligned} \quad (\text{C.38})$$

$$d_{12}^{(4)} = \frac{(M + 2m_f)}{4M} \quad (\text{C.39})$$

$$d_{22}^{(4)} = \frac{1}{2} \quad (\text{C.40})$$

$$d_{11}^{(5)} = \frac{(M + 2m_f)^2(-1 + z)(1 + vz')}{2(M + 2m_f - Mz)^2 + 8\gamma} \quad (\text{C.41})$$

$$\begin{aligned} d_{12}^{(5)} = & \frac{(M + 2m_f)(1 + vz')}{4((M + 2m_f - Mz)^2 + 4\gamma)} \left[(-4m_f v(-1 + z)z' - 4\frac{m_f^2}{M}(z - vz') \right. \\ & \left. + M(-1 + z)(-z + v(-1 + z)z') + \frac{4}{M}(-1 + vz')\gamma) \right] \end{aligned} \quad (\text{C.42})$$

$$d_{21}^{(5)} = \frac{M(M + 2m_f)(-1 + z)(1 + vz')}{(M + 2m_f - Mz)^2 + 4\gamma} \quad (\text{C.43})$$

$$\begin{aligned} d_{22}^{(5)} = & \frac{1}{2(M + 2m_f - Mz)^2 + 8\gamma} [((1 + vz')(-4Mm_f v(-1 + z)z' - 4m_f^2(z - vz') \\ & + M^2(-1 + z)(-z + v(-1 + z)z') + 4(-1 + vz')\gamma))] \end{aligned} \quad (\text{C.44})$$

$$d_{11}^{(6)} = \frac{(2M(M + 2m_f)(-1 + v)(-1 + z))}{((M + 2m_f - Mz)^2 + 4\gamma)} \quad (\text{C.45})$$

$$d_{12}^{(6)} = \frac{1}{(M((M + 2m_f - Mz)^2 + 4\gamma))} \left[(M + 2m_f)(-1 + v)(M^2(v(-1 + z) - z)(-1 + z) + 2Mm_f(-1 - 2v(-1 + z) + z) + 4(-1 + v)(m_f^2 + \gamma)) \right] \quad (\text{C.46})$$

$$d_{21}^{(6)} = \frac{(4M^2(-1 + v)(-1 + z))}{(M + 2m_f - Mz)^2 + 4\gamma} \quad (\text{C.47})$$

$$d_{22}^{(6)} = \frac{1}{(M + 2m_f - Mz)^2 + 4\gamma} \left[2(-1 + v)(M^2(v(-1 + z) - z)(-1 + z) + 2Mm_f(-1 - 2v(-1 + z) + z) + 4(-1 + v)(m_f^2 + \gamma)) \right] \quad (\text{C.48})$$